

HW 3 Other Problems (Rapidity Part 2 and Others) [FINALIZED]

E1) Explain why for beam particles no matter which situation or if you define rapidity by the more common textbook definition or the slides definition , the factor by which time is dilated in the beam particles frame of reference is equal to $\cosh y_{\text{beam}}$.

E2) a) Use the Jacobian function $dy/d\eta$ for rewriting dN/dy in terms of $dN/d\eta$ for constant or fixed p_T as a function of η or y . You may use the result in class, but derive it again $\rightarrow$ at fixed p_T , $dy/d\eta = \beta$.

b) Plot this function at least in the vicinity of $y=0$ (you may either turn in a plot generated with a computer, or just draw what it looks like after plotting it on a computer).

c) Also evaluate the difference numerically between η and y for either protons and kaons (your choice) at a three integer absolute values of rapidity other than 0. Possibly use the last relation in part a) or just evaluate it from the formulas.

(Hints: comparing y and η (pseudorap), but setting $y = 1, 2, 3$, can only determine ratio of p_T/E (p_z/E) ...must specify m , AND p_T to fix p_z , $|p|$, and $E \rightarrow$ from $p_z/p = \cos\theta$ (determine θ) calc η)

E3.) (a) Find the shape (functional form up to an overall constant) of the distribution dn/dE_γ of the energies of decay photons in the lab frame for a $\pi^0 \rightarrow 2\gamma$ decay where the π^0 's decaying all have highly relativistic energy E_{π^0} and momenta directed along the same direction. (you may presume without justification that the min energy is 0 and and the max lab energy for the photons is E_{π^0} . –recall that the decay angles in the center of mass are such that $dN/d\cos\theta^*$ distribution is flat i.e. $\propto$ a constant. Read the other parts below for more hints starting this part)

b) Find the shape of the lab frame θ or $\cos\theta$ distribution ($dn/d\theta$ or $dN/d(\cos\theta)$) for the same case as part (a), except this time the direction of all π^0 momenta are in the positive z -direction. Plot this function as a function of angle for 1 GeV and 3 GeV π^0 's. (Hint: this time, evaluate $d(\cos\theta^*)/d(\cos\theta)$ from the transformation equations)

c) Plot the function derived in the slides for the shape of the distribution $dn/d\Delta\theta$ of **opening** angles between the two photons for 1 GeV π^0 's. This angle is sometimes also referred to as θ_{12} and in the slides w/ the formula, note that θ was actually redefined to be $\Delta\theta$. This angle is NOT simply the difference of each photon's polar angle θ being used in part b. You do not need to confirm the derivation of that function, but it's not hard using the ingredients in the slides.

Lancaster and Blundell Problems:

Equation numbers refer to Lancaster and Blundell:

L&B 1) a) Show that $\rho(x)$ as defined in Eq 6.12 allows us to write 6.10.

b) Using the relativistic free particle wave function $\phi(x) = N e^{-ip^\mu x_\mu}$ to evaluate 6.13, show the explicit form of $j_\mu(x)$ which includes 6.14.

c) Explain in a sentence or two why 6.14 is problematic

L&B 2) a) Write the mode expansion for the real scalar field found in Chapter 11, and explain briefly the connection to 6.19 i.e. what the two terms represent.

b) Explain how Lancaster Blundell explains the factor of $1/\sqrt{2E}$ in the mode expansion and corresponding related norm factors in four-momentum eigenstates, and annihilation/creation operators.

c) Describe what happens to any constants or normalization factors involving E (same E as in part b of this problem), in each of the main steps in deriving the form of the Hamilton field operator, Eq 11.38, essentially how the factor of E ends up in that expression how and where it does. You don't need to write out the entire expressions, nor show all the algebra needed to get from step to step, but rather you can summarize what's done in each step e.g. even mostly just in words, only highlighting what happens to in factors involving E that lead to the energy dependence of 11.38.